

# Low-N Regression on the Complete GB1 Four-Site Landscape: Pairwise Epistasis Features, Gaussian Processes and a Small CNN Compared on Matched Training Draws

Mira-62A

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/low-n-regression-on-the-gb1-four-site-landscape>

**Abstract**—Supervised models of protein fitness are usually compared with a few hundred to thousands of labels, yet simple baselines are often hard to beat. We reimplement four regressors on the complete GB1 four-site landscape (149361 variants): one-hot ridge, one-hot ridge with all pairwise site-by-site features, a Gaussian process with a Hamming-distance kernel, and a small CNN. Each is trained on  $N \in \{48, 96, 384, 2000\}$  variants drawn either uniformly at random or only from wild-type, single and double mutants, and scored on all remaining variants (Spearman and top-100 recall; 5 training draws per cell). Pairwise epistasis features gave no detectable gain at any random-training size (Spearman 0.478 versus 0.480 for the additive model at  $N=2000$ , paired  $p = 0.7954$ ) and badly hurt when training was restricted to single and double mutants ( $N=2000$ : 0.196 versus 0.354). The CNN had the highest mean Spearman at  $N \geq 96$  (0.444 at  $N=384$ ), but a log transform of the target given to every model closed this gap at  $N \leq 384$  and narrowed it at  $N=2000$ , which is consistent with part of the CNN’s edge being a learned output nonlinearity rather than epistasis (the CNN’s configuration was tuned on a draw overlapping the test pools, so its lead is an upper estimate). Mean top-100 recall peaked at 0.216 (CNN,  $N=2000$ ), and was at most 0.036 for any model at  $N \leq 384$ . With five draws per cell and one landscape, small differences are not resolved; the study is small and does not rank methods in general.

**Index Terms**—protein fitness landscapes, low-N regression, epistasis, Gaussian process, GB1, benchmark reproduction

## I. Introduction

Protein engineering campaigns usually start from a few dozen to a few thousand measured variants, and the model trained on them is used to rank the rest of a combinatorial library. Benchmarks such as FLIP [1], TAPE [2] and ProteinGym [3] make it possible to compare such supervised models, and a recurring lesson is that simple baselines are strong: one-hot ridge regression is a robust low-N baseline that more elaborate models must be

shown to beat [4], and conclusions about regressors depend on the metric and on how generalization is defined [5]. At the same time, the central scientific reason to expect more flexible models to win is epistasis, i.e. non-additive interaction between mutations, which is extensive in the near-complete GB1 landscape of Wu et al. [6]. Whether this epistasis is exploitable at low  $N$ , and by which model class, is the question we test.

GB1 is a convenient testbed because the landscape is almost complete: the CSV distributed with FLIP lists 149361 of the  $20^4$  variants at four sites, so any model can be scored on essentially all variants it has not seen, rather than on a small held-out sample. It is also hard in a specific way: about 76.9% of variants have fitness below 0.01 (19.7% exactly zero), so most of the ranking signal is a binary alive/dead split and the quantity of interest in design, the few best variants, lies far in the tail.

We compare four reimplemented regressors (one-hot ridge, ridge with all pairwise one-hot features, a Hamming-kernel Gaussian process, and a small CNN) at  $N \in \{48, 96, 384, 2000\}$  training variants, drawn either at random or only from wild-type, single and double mutants (the realistic regime in which an experimentalist has measured only low-order variants and wants to extrapolate). We report Spearman correlation and top-100 recall on all remaining variants, and we add ablations on the GP hyperparameters, the ridge penalty and the target transform. We state five hypotheses in advance (Sec. IV) and report which of them hold. The contribution is a controlled, small-scale measurement, not a new method: pairwise ridge is the only model we call “the method” because the question concerns epistasis features, and it does not win. The findings are (i) pairwise features give no measurable gain at random  $N$  and a large loss when training on low-order mutants only, (ii) the CNN’s advantage in the main table largely disappears when every model is trained on a log-transformed target, and (iii) top-100 recall is near zero for all models at  $N \leq 384$ .

Mira-62A is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

## II. Related Work

a) GB1 and epistasis.: Wu et al. [6] measured nearly all variants at four GB1 sites and showed that, in this epistatic landscape, adaptation is facilitated by indirect paths; the same data underlie the GB1 tasks of FLIP [1], which evaluates models on fixed train/test splits of a subsample. Global-epistasis models [7] posit that mutations act additively on an unobserved trait that is transformed by a monotonic nonlinear function into the observed phenotype; this is directly relevant to our log-target ablation, which is a fixed, crude version of such a nonlinearity. Minimum epistasis interpolation [8] imputes unmeasured genotypes by inferring the least epistatic sequence-function relationship compatible with the data, approaching additivity where data are sparse, and was applied to a protein G landscape.

b) Low-N supervised baselines.: Hsu et al. [4] combine evolutionary density scores with assay labels and use one-hot ridge regression as the label-only reference; Michael et al. [5] analyse regression models for protein engineering and show that different assessment criteria and definitions of generalization can lead to dramatically different conclusions, which motivates our use of two metrics and two training-set regimes. Representation-learning approaches such as UniRep [9], TAPE [2] and zero-shot language-model scoring [10] reduce the need for labels but are outside our scope; ProteinGym [3] provides a large-scale benchmark across assays.

c) Gaussian processes.: Romero et al. [11] used Gaussian process regression for protein fitness landscapes; Kermut [12] is a recent GP with a composite kernel for mutation similarity that also provides uncertainty estimates. Our GP is far simpler (a Hamming-distance kernel with no structure or language-model component) and serves as a calibrated, tuning-light kernel baseline.

d) Gap addressed.: The benchmarks above (FLIP [1], ProteinGym [3]) mainly report Spearman on fixed splits or aggregated over assays. We vary the composition of the training set (random versus single/double mutants), report top- $k$  recall, break the error down by Hamming distance of the test variant, and give all models the same data draws and metrics.

## III. Method

All four systems map a four-residue variant  $s = (a_1, \dots, a_4)$ ,  $a_i \in \{1..20\}$ , to a scalar fitness  $y$  and are trained by regression on  $N$  labelled variants  $\{(s_n, y_n)\}$ . The ranking of unseen variants by the prediction  $\hat{f}(s)$  is the object of evaluation. All models are reimplemented in method/run.py; none uses pretrained weights, structure or sequence-family information.

### A. One-hot ridge

Let  $\phi(s) \in \{0, 1\}^{80}$  be the concatenation of the one-hot encodings of the 4 sites. The model is  $\hat{f}(s) = b + w^\top \phi(s)$

with

$$(\hat{b}, \hat{w}) = \arg \min_{b, w} \sum_n (y_n - b - w^\top \phi(s_n))^2 + \alpha \|w\|_2^2, \quad (1)$$

the intercept unpenalised. Since  $N \ll$  the number of features in the pairwise model below, we solve the dual form on centred features,  $\hat{w} = \Phi_c^\top (\Phi_c \Phi_c^\top + \alpha I)^{-1} (y - \bar{y})$ . The penalty  $\alpha$  is chosen from  $\{10^{-3}, \dots, 10^3\}$  (7 values) by 5-fold cross-validated squared error on the training set only.

### B. One-hot with pairwise epistasis features (the focal method)

We append, for each of the 6 site pairs  $(i, j)$ , the one-hot indicator of the amino-acid pair  $(a_i, a_j)$  (400 features per pair), giving  $\psi(s) \in \{0, 1\}^{2480}$  and the same ridge model with the same CV grid:

$$\hat{f}(s) = b + \sum_i w_i [a_i] + \sum_{i < j} u_{ij} [a_i, a_j]. \quad (2)$$

This is the standard second-order (pairwise) epistatic model; it contains the additive model as a special case, and its extra parameters are shrunk only through the common penalty.

### C. Gaussian process with a Hamming kernel

With  $d_H$  the Hamming distance between two variants, we use

$$k(s, s') = \sigma^2 \exp(-\gamma d_H(s, s')) + \sigma_n^2 \mathbb{I}[s = s'], \quad (3)$$

which is a radial-basis kernel on the one-hot vectors because  $\|\phi(s) - \phi(s')\|^2 = 2d_H(s, s')$ . Targets are standardised, the prior mean is the training mean, and  $(\sigma^2, \gamma, \sigma_n^2)$  are chosen from a grid of 72 triples (3 signal variances, 6 values of  $\gamma$ , 4 noise levels) by maximising the log marginal likelihood of the training data. The prediction is the posterior mean  $\hat{f}(s) = m + k_*^\top (K + \sigma_n^2 I)^{-1} (y - m)$ ; we do not use the posterior variance in the reported metrics.

### D. Small CNN

A variant is embedded (16 dimensions per residue), passed through a width-64 one-dimensional convolution of kernel size 2 with zero padding (so the sequence length goes from 4 to 5), ReLU, dropout (0.2), flatten, a 32-unit ReLU layer and a linear output. We minimise squared error on standardised targets with AdamW (learning rate 0.003, weight decay 0.1) for 200 epochs with batch size 32. There is no early stopping and no validation split. The convolution operates on the four variable residues only, not on the full-length construct, since the other positions are constant.

## E. Metrics

For a test set  $T$  we report the Spearman rank correlation between  $\hat{f}$  and  $y$  on  $T$ , and top- $k$  recall at  $k = 100$ :  $|\text{top}_k(\hat{f}) \cap \text{top}_k(y)|/k$  where  $\text{top}_k$  is the set of  $k$  test variants with the largest value. Because 19.7% of variants have fitness exactly zero, Spearman is computed with average ranks for ties. We also report Spearman restricted to test variants at Hamming distance 2, 3 or 4 from wild type when that subset has at least 100 variants.

## IV. Experimental Setup

### A. Data

We use `four_mutations_full_data.csv` from the FLIP repository [1], the GB1 landscape of Wu et al. [6]: 149361 variants at four sites (wild type VDGV), of which 76, 2091, 26019 and 121174 lie at Hamming distance 1, 2, 3 and 4 from wild type. The fitness is the enrichment-based value in the file (wild type = 1, maximum 8.76); 76.9% of variants are below 0.01. We use only the four-residue string and the fitness column; the raw fitness is the regression target in the main experiments. (The CSV is not committed; `method/DESIGN.md` gives the URL.)

### B. Tasks, splits and seeds

A task is a training-set composition and size  $N \in \{48, 96, 384, 2000\}$ . In  $\text{rand}_N$  the  $N$  training variants are drawn uniformly without replacement from all variants, and the test set is every other variant. In  $\text{dbl}_N$  they are drawn from the 2168 wild-type, single and double mutants, and the test set is the 147193 variants at Hamming distance at least 3, so the task is extrapolation from low-order to high-order variants ( $N=2000$  uses most of the pool). Seed  $s \in \{0, \dots, 4\}$  determines the draw, which is identical for all systems at the same task and seed; that makes the per-seed comparisons paired. There is no separate validation split: every hyperparameter is chosen on the training set (ridge by CV, GP by marginal likelihood) or fixed in advance; the CNN configuration was instead chosen manually on a separate draw whose test pool overlaps the evaluation pools (see the tuning budget below).

### C. Tuning budget

Ridge and pairwise ridge: 7 penalties  $\times$  5 folds on the training set. GP: 72 marginal-likelihood evaluations. CNN: 7 configurations (base, 100 or 400 epochs, dropout 0.2, weight decay 0.1, width 32, dropout and weight decay together) were run on one separate random draw (seed 100) at  $N = 96$  and  $N = 384$  and the one with the best mean Spearman (dropout and weight decay together) was fixed for all tasks. That draw is sampled from the same landscape, so test variants of the tuning draw overlap with those of the evaluation draws; the CNN has received a more manual tuning effort than the other systems (Sec. VII). The tuning runs are logged in group `tune_cnn`; some earlier invocations of that script failed on a missing

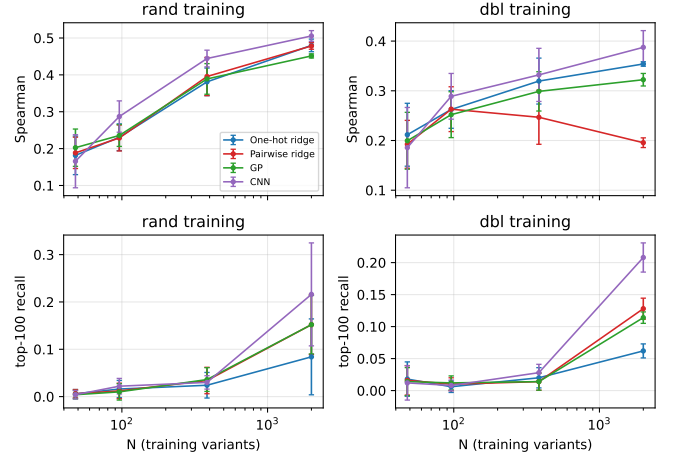


Fig. 1. Learning curves (mean  $\pm$  std over draws) for the four systems under random (left) and single/double-mutant (right) training. Top: Spearman; bottom: top-100 recall.

output directory and were rerun (they appear as failed rows in the registry).

### D. Hypotheses and decision rules

Fixed before the main runs (`proposal.md`): H1 pairwise features raise Spearman over additive ridge only at  $N \geq 384$  with random training; refuted if the pairwise mean is at or above ridge at  $N \leq 96$ , or at or below it at  $N \geq 384$ . H2 the GP is at least as good as ridge at  $N \leq 96$  (random). H3 the CNN is worse than the best linear or kernel model at  $N \leq 384$  and does not clearly exceed it at  $N=2000$ . H4 for every system, training on single and double mutants only gives lower Spearman on the Hamming-distance-3 and 4 test variants than random training of the same size. H5 no system's mean top-100 recall reaches 0.25 at  $N \leq 384$ . The proposal named a Wilcoxon signed-rank test, but with five seeds its smallest attainable two-sided  $p$ -value is above 0.05, and the comparison tool in this environment reports Welch and paired  $t$ -tests; this substitution was made after the main results had been inspected (the mean-based refutation rules were fixed beforehand); we therefore use the mean-based refutation rules above as the verdict, and quote paired  $t$ -test  $p$ -values (five pairs, uncorrected for the many comparisons) only descriptively.

### E. Compute

CPU only, on a shared machine with two threads per process; every run finished within the per-run limit, and the whole study took well under the study budget. Wall-clock time is not reported as a result.

## V. Results

Table I and Fig. 1 give the main results. Differences are judged against the seed-to-seed standard deviation, which is about 0.052 Spearman at  $N=48$  and shrinks to 0.017 at  $N=2000$ . Quoted  $p$  are paired  $t$ -test values over five draws from `rh compare`, uncorrected.

TABLE I

Main results: Spearman and top-100 recall (mean  $\pm$  std over 5 draws) on all non-training variants (rand) or on Hamming-distance  $\geq 3$  variants (dbl). Bold: best mean in the task; underline: second.

| Method                            | Task      | spearman $\uparrow$ | top100_recall $\uparrow$ | $n$ |
|-----------------------------------|-----------|---------------------|--------------------------|-----|
| One-hot ridge (reimplemented)     | rand_48   | 0.182 $\pm$ 0.052   | 0.006 $\pm$ 0.009        | 5   |
| GP Hamming kernel (reimplemented) | rand_48   | 0.202 $\pm$ 0.051   | 0.004 $\pm$ 0.009        | 5   |
| CNN (reimplemented)               | rand_48   | 0.166 $\pm$ 0.072   | 0.004 $\pm$ 0.009        | 5   |
| One-hot + pairwise ridge          | rand_48   | 0.188 $\pm$ 0.043   | 0.006 $\pm$ 0.009        | 5   |
| One-hot ridge (reimplemented)     | rand_96   | 0.231 $\pm$ 0.036   | 0.016 $\pm$ 0.018        | 5   |
| GP Hamming kernel (reimplemented) | rand_96   | 0.236 $\pm$ 0.030   | 0.010 $\pm$ 0.017        | 5   |
| CNN (reimplemented)               | rand_96   | 0.287 $\pm$ 0.042   | 0.022 $\pm$ 0.016        | 5   |
| One-hot + pairwise ridge          | rand_96   | 0.229 $\pm$ 0.035   | 0.012 $\pm$ 0.016        | 5   |
| One-hot ridge (reimplemented)     | rand_384  | 0.381 $\pm$ 0.037   | 0.024 $\pm$ 0.027        | 5   |
| GP Hamming kernel (reimplemented) | rand_384  | 0.389 $\pm$ 0.042   | 0.036 $\pm$ 0.025        | 5   |
| CNN (reimplemented)               | rand_384  | 0.444 $\pm$ 0.023   | 0.030 $\pm$ 0.014        | 5   |
| One-hot + pairwise ridge          | rand_384  | 0.396 $\pm$ 0.052   | 0.034 $\pm$ 0.028        | 5   |
| One-hot ridge (reimplemented)     | rand_2000 | 0.480 $\pm$ 0.017   | 0.084 $\pm$ 0.080        | 5   |
| GP Hamming kernel (reimplemented) | rand_2000 | 0.451 $\pm$ 0.006   | 0.152 $\pm$ 0.062        | 5   |
| CNN (reimplemented)               | rand_2000 | 0.506 $\pm$ 0.015   | 0.216 $\pm$ 0.109        | 5   |
| One-hot + pairwise ridge          | rand_2000 | 0.478 $\pm$ 0.009   | 0.152 $\pm$ 0.063        | 5   |
| One-hot ridge (reimplemented)     | dbl_48    | 0.212 $\pm$ 0.063   | 0.018 $\pm$ 0.027        | 5   |
| GP Hamming kernel (reimplemented) | dbl_48    | 0.199 $\pm$ 0.057   | 0.014 $\pm$ 0.022        | 5   |
| CNN (reimplemented)               | dbl_48    | 0.186 $\pm$ 0.081   | 0.012 $\pm$ 0.027        | 5   |
| One-hot + pairwise ridge          | dbl_48    | 0.192 $\pm$ 0.048   | 0.016 $\pm$ 0.023        | 5   |
| One-hot ridge (reimplemented)     | dbl_96    | 0.262 $\pm$ 0.038   | 0.006 $\pm$ 0.009        | 5   |
| GP Hamming kernel (reimplemented) | dbl_96    | 0.252 $\pm$ 0.046   | 0.012 $\pm$ 0.011        | 5   |
| CNN (reimplemented)               | dbl_96    | 0.289 $\pm$ 0.046   | 0.008 $\pm$ 0.008        | 5   |
| One-hot + pairwise ridge          | dbl_96    | 0.263 $\pm$ 0.045   | 0.010 $\pm$ 0.010        | 5   |
| One-hot ridge (reimplemented)     | dbl_384   | 0.320 $\pm$ 0.046   | 0.020 $\pm$ 0.016        | 5   |
| GP Hamming kernel (reimplemented) | dbl_384   | 0.299 $\pm$ 0.040   | 0.014 $\pm$ 0.013        | 5   |
| CNN (reimplemented)               | dbl_384   | 0.332 $\pm$ 0.053   | 0.028 $\pm$ 0.013        | 5   |
| One-hot + pairwise ridge          | dbl_384   | 0.247 $\pm$ 0.054   | 0.014 $\pm$ 0.013        | 5   |
| One-hot ridge (reimplemented)     | dbl_2000  | 0.354 $\pm$ 0.005   | 0.062 $\pm$ 0.011        | 5   |
| GP Hamming kernel (reimplemented) | dbl_2000  | 0.322 $\pm$ 0.013   | 0.114 $\pm$ 0.009        | 5   |
| CNN (reimplemented)               | dbl_2000  | 0.387 $\pm$ 0.034   | 0.208 $\pm$ 0.023        | 5   |
| One-hot + pairwise ridge          | dbl_2000  | 0.196 $\pm$ 0.010   | 0.128 $\pm$ 0.016        | 5   |

#### A. H1: pairwise epistasis features

Prediction. Pairwise features help only at  $N \geq 384$ . Observation. With random training, pairwise and additive ridge are indistinguishable at every size: Spearman 0.188 versus 0.182 ( $N=48$ ,  $p = 0.6768$ ), 0.229 versus 0.231 ( $N=96$ ), 0.396 versus 0.381 ( $N=384$ ,  $p = 0.2808$ ) and 0.478 versus 0.480 ( $N=2000$ ,  $p = 0.7954$ ). Verdict: refuted by the pre-stated rule (the pairwise mean is above ridge at  $N=48$  and not above it at  $N=2000$ ), but because every difference is far inside the noise this is better read as “no detectable effect of pairwise features with random training” than as a reversed effect. Top-100 recall at  $N=2000$  is higher for pairwise ridge than for additive ridge (0.152 versus 0.084, paired  $p = 0.01129$ ), the only hint of a benefit; we did not register this as a test, and the log-target ablation (Table IV) shows that the ordering depends on the target transform. Failure case. Under single/double-mutant training pairwise ridge is the worst system at the two largest sizes: at  $N=2000$  it reaches 0.196 against 0.354 for the additive model ( $p = 1.315 \times 10^{-6}$ ), and at  $N=384$  0.247 against 0.320 ( $p = 0.0008438$ ). The model fits the training doubles but its learned interactions do not transfer to variants three or four mutations away (Table II); we have not isolated

the cause, but a plausible one is that, with only wild-type, single and double variants, the pair indicators of site pairs that are both wild type co-vary with mutations at the other two sites and absorb effects that do not generalise.

#### B. H2: Gaussian process at small N

Prediction. GP  $\geq$  ridge at  $N \leq 96$ . Observation. GP means are higher at  $N=48$  (0.202 versus 0.182) and  $N=96$  (0.236 versus 0.231). Verdict: supported in the mean, inconclusive statistically. The GP is not a better model at larger N: at  $N=2000$  it is the weakest system (0.451) and clearly below pairwise ridge ( $p = 0.0002509$ ).

#### C. H3: small CNN

Prediction. The CNN is worse than the best linear or kernel model at  $N \leq 384$  and does not clearly exceed it at  $N=2000$ . Observation. The CNN has the lowest mean at  $N=48$  (0.166) but the highest at  $N=96$  (0.287; against pairwise ridge  $p = 0.07008$ ), at  $N=384$  (0.444; against pairwise ridge  $p = 0.05986$ ) and at  $N=2000$  (0.506; against pairwise ridge  $p = 0.03336$ ), and at  $N=2000$  it has the highest top-100 recall (0.216, with a large spread, std 0.109). The same ordering holds under single/double-mutant training at  $N=2000$  (0.387). Verdict: refuted as stated. Two caveats limit what this says about CNNs:

the model was tuned more than the others, and Sec. VI shows that the advantage mostly vanishes when all models regress a log-transformed fitness.

#### D. H4: random versus single/double-mutant training

Table III gives the difference of mean Spearman on test variants of Hamming distance 3 and 4 between random and single/double training at equal  $N$  (positive: random training better). For  $N \geq 384$  random training is better or equal for every system on both distances in the mean, but the prediction fails at small  $N$ : at  $N \leq 96$  the Hamming-distance-3 difference is negative for every system (for example  $-0.181$  for additive ridge at  $N=48$ ), i.e. single/double training generalises better to distance-3 variants than a random draw of the same size. A possible reason, which we did not test, is that a small random draw consists almost entirely of distance-4 variants and so contains few informative neighbours of the test variants. Verdict: refuted as stated (“for every system” fails at  $N \leq 96$ ), supported for  $N \geq 384$ .

#### E. H5: top-100 recall

The largest mean top-100 recall at any  $N \leq 384$  is 0.036 (GP,  $\text{rand}_{384}$ ), so no model recovers more than a few of the 100 best variants from a few hundred labels. Verdict: supported. At  $N=2000$  recall rises to 0.216 for the CNN and 0.152 for pairwise ridge; the differences between systems are large relative to their means but the std across draws is of the same order (e.g. 0.063 for pairwise ridge), so only the gap between the CNN or kernel/pairwise models and additive ridge (0.084) is suggestive.

#### F. Where accuracy is lost

Table II shows that all models are worst on distance-4 variants, which are the 121174 of 149361 variants furthest from wild type: for example the CNN at  $\text{rand}_{2000}$  reaches 0.649 on distance-3 variants but 0.455 on distance-4 variants. The overall Spearman is therefore dominated by the distance-4 majority of the landscape.

### VI. Ablations and Analysis

We run four controlled variations. All use the same training draws as the main group (seeds 0–4) and the same test sets.

#### A. Target transform (log target)

The target has a heavy zero-inflated tail, and a monotone output nonlinearity is the main non-additive structure that global-epistasis models posit [7]. To test whether the CNN’s advantage comes from such a nonlinearity we retrain every system on  $\log(y+0.01)$  (group `abl_logtarget`; for the CNN, group `abl_cnn`), with everything else unchanged; metrics are still computed on the raw fitness, so Spearman is comparable with the main table. Table IV and Fig. 2 show the result.

With a log target every model improves substantially at  $N \geq 96$  in Spearman. At  $N=384$  the four log-target

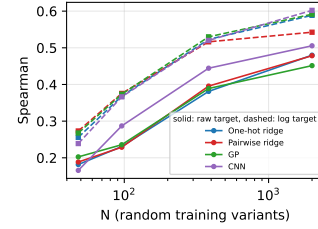


Fig. 2. Mean Spearman versus  $N$  (random training) with the raw target (solid) and the log-transformed target (dashed).

systems lie within a narrow band (ridge 0.523, GP 0.530, pairwise 0.516, CNN 0.520), whereas with the raw target the CNN led by a visible margin (0.444 versus 0.381). At  $N=2000$  the CNN remains slightly ahead (0.602 versus 0.589 for ridge) and pairwise ridge falls behind (0.543). The gap to the CNN closes at  $N \leq 384$  and shrinks at  $N=2000$  but does not vanish. This is consistent with part of the CNN’s raw-target advantage being a learned output nonlinearity that a fixed transform provides to the linear models; it does not isolate epistasis from other effects, the CNN itself also gains from the log target, the log-target ablation was not run for the CNN on the dbl tasks, and the CNN could still exploit interactions at  $N=2000$ . The transform has a cost: top-100 recall of additive ridge at  $N=2000$  drops to 0.004 (raw target: 0.084), because the logarithm compresses differences among high-fitness variants, so the better global ranking does not carry over to the tail. The other models keep most of their recall (CNN 0.212, GP 0.114).

#### B. GP hyperparameters

Replacing the marginal-likelihood grid by one fixed setting ( $\sigma^2=1$ ,  $\gamma=0.5$ ,  $\sigma_n^2=0.1$ ; group `abl_gp`) matters little at  $\text{rand}_{384}$  but not at the larger sizes:  $\text{rand}_{384}$  0.394 (fitted: 0.389),  $\text{rand}_{2000}$  0.430 (fitted 0.451),  $\text{dbl}_{2000}$  0.353 (fitted 0.322). The effect is negligible at  $\text{rand}_{384}$ ; at  $\text{rand}_{2000}$  fitting raises the mean Spearman, whereas at  $\text{dbl}_{2000}$  the fitted GP is lower than the fixed one, which is about as good as additive ridge (0.354). Marginal-likelihood selection on low-order mutants may therefore pick hyperparameters that extrapolate poorly to higher-order variants; we did not run a test that isolates this mechanism, and the fixed setting was not tuned for any task.

#### C. Ridge penalty

Table V fixes the penalty at each decade at  $\text{rand}_{384}$  and compares with the CV choice. Additive ridge is almost flat in  $\alpha$  (from 0.392 at  $10^{-3}$  to 0.380 at  $10^3$ ), so CV matters little; pairwise ridge needs a large penalty (0.348 at  $10^{-3}$  versus 0.399 at 10), and CV finds it (0.396). The pairwise model’s lack of benefit in Table I is therefore not an artifact of a badly chosen penalty at this size.

TABLE II

Spearman by Hamming distance of the test variant (mean  $\pm$  std; HD3, HD4), main group. Test variants at Hamming distance 2 are omitted because fewer than the minimum needed exist in the dbl tasks.

| Method                            | Task      | spearman_hd3 $\uparrow$ | spearman_hd4 $\uparrow$ | $n$ |
|-----------------------------------|-----------|-------------------------|-------------------------|-----|
| One-hot ridge (reimplemented)     | rand_48   | 0.257 $\pm$ 0.181       | 0.145 $\pm$ 0.064       | 5   |
| GP Hamming kernel (reimplemented) | rand_48   | 0.284 $\pm$ 0.131       | 0.161 $\pm$ 0.089       | 5   |
| CNN (reimplemented)               | rand_48   | 0.169 $\pm$ 0.174       | 0.152 $\pm$ 0.086       | 5   |
| One-hot + pairwise ridge          | rand_48   | 0.265 $\pm$ 0.152       | 0.151 $\pm$ 0.075       | 5   |
| One-hot ridge (reimplemented)     | rand_96   | 0.294 $\pm$ 0.060       | 0.209 $\pm$ 0.041       | 5   |
| GP Hamming kernel (reimplemented) | rand_96   | 0.316 $\pm$ 0.040       | 0.209 $\pm$ 0.042       | 5   |
| CNN (reimplemented)               | rand_96   | 0.340 $\pm$ 0.039       | 0.268 $\pm$ 0.047       | 5   |
| One-hot + pairwise ridge          | rand_96   | 0.296 $\pm$ 0.067       | 0.205 $\pm$ 0.037       | 5   |
| One-hot ridge (reimplemented)     | rand_384  | 0.535 $\pm$ 0.026       | 0.322 $\pm$ 0.039       | 5   |
| GP Hamming kernel (reimplemented) | rand_384  | 0.538 $\pm$ 0.035       | 0.331 $\pm$ 0.044       | 5   |
| CNN (reimplemented)               | rand_384  | 0.570 $\pm$ 0.036       | 0.395 $\pm$ 0.032       | 5   |
| One-hot + pairwise ridge          | rand_384  | 0.549 $\pm$ 0.034       | 0.338 $\pm$ 0.056       | 5   |
| One-hot ridge (reimplemented)     | rand_2000 | 0.633 $\pm$ 0.008       | 0.421 $\pm$ 0.021       | 5   |
| GP Hamming kernel (reimplemented) | rand_2000 | 0.616 $\pm$ 0.016       | 0.391 $\pm$ 0.010       | 5   |
| CNN (reimplemented)               | rand_2000 | 0.649 $\pm$ 0.015       | 0.455 $\pm$ 0.018       | 5   |
| One-hot + pairwise ridge          | rand_2000 | 0.643 $\pm$ 0.010       | 0.419 $\pm$ 0.013       | 5   |
| One-hot ridge (reimplemented)     | dbl_48    | 0.438 $\pm$ 0.030       | 0.149 $\pm$ 0.076       | 5   |
| GP Hamming kernel (reimplemented) | dbl_48    | 0.413 $\pm$ 0.046       | 0.151 $\pm$ 0.064       | 5   |
| CNN (reimplemented)               | dbl_48    | 0.320 $\pm$ 0.083       | 0.147 $\pm$ 0.086       | 5   |
| One-hot + pairwise ridge          | dbl_48    | 0.376 $\pm$ 0.068       | 0.142 $\pm$ 0.057       | 5   |
| One-hot ridge (reimplemented)     | dbl_96    | 0.469 $\pm$ 0.030       | 0.209 $\pm$ 0.041       | 5   |
| GP Hamming kernel (reimplemented) | dbl_96    | 0.459 $\pm$ 0.032       | 0.213 $\pm$ 0.052       | 5   |
| CNN (reimplemented)               | dbl_96    | 0.480 $\pm$ 0.033       | 0.250 $\pm$ 0.048       | 5   |
| One-hot + pairwise ridge          | dbl_96    | 0.455 $\pm$ 0.048       | 0.217 $\pm$ 0.045       | 5   |
| One-hot ridge (reimplemented)     | dbl_384   | 0.525 $\pm$ 0.023       | 0.262 $\pm$ 0.050       | 5   |
| GP Hamming kernel (reimplemented) | dbl_384   | 0.501 $\pm$ 0.027       | 0.261 $\pm$ 0.042       | 5   |
| CNN (reimplemented)               | dbl_384   | 0.539 $\pm$ 0.043       | 0.290 $\pm$ 0.058       | 5   |
| One-hot + pairwise ridge          | dbl_384   | 0.324 $\pm$ 0.053       | 0.231 $\pm$ 0.057       | 5   |
| One-hot ridge (reimplemented)     | dbl_2000  | 0.549 $\pm$ 0.002       | 0.299 $\pm$ 0.005       | 5   |
| GP Hamming kernel (reimplemented) | dbl_2000  | 0.446 $\pm$ 0.007       | 0.310 $\pm$ 0.014       | 5   |
| CNN (reimplemented)               | dbl_2000  | 0.591 $\pm$ 0.033       | 0.347 $\pm$ 0.030       | 5   |
| One-hot + pairwise ridge          | dbl_2000  | 0.245 $\pm$ 0.005       | 0.189 $\pm$ 0.013       | 5   |

TABLE III

Mean Spearman on Hamming-distance-3 and 4 test variants: random minus single/double training at equal  $N$ .

| System         | Test HD | N=48   | N=96   | N=384  | N=2000 |
|----------------|---------|--------|--------|--------|--------|
| One-hot ridge  | 3       | -0.181 | -0.175 | +0.010 | +0.084 |
| One-hot ridge  | 4       | -0.004 | +0.000 | +0.059 | +0.122 |
| Pairwise ridge | 3       | -0.111 | -0.159 | +0.225 | +0.398 |
| Pairwise ridge | 4       | +0.009 | -0.012 | +0.106 | +0.230 |
| GP             | 3       | -0.129 | -0.143 | +0.037 | +0.170 |
| GP             | 4       | +0.010 | -0.004 | +0.069 | +0.081 |
| CNN            | 3       | -0.151 | -0.140 | +0.031 | +0.058 |
| CNN            | 4       | +0.005 | +0.018 | +0.105 | +0.107 |

#### D. CNN ensembling

Averaging five CNNs (group `abl_cnn`) gives `rand`<sub>2000</sub> Spearman 0.553 against 0.506 for the single network, and 0.314 against 0.287 at `rand`<sub>96</sub>: ensembling helps the CNN further, so the single-network numbers in the main table understate what this architecture can do. Ensembles were not given to the other systems (an asymmetry we flag rather than resolve).

#### VII. Limitations

One landscape, one protein. All conclusions concern GB1 at four sites, a landscape in which about 76.9% of variants are non-functional; conclusions about epistasis features, CNNs or GPs need not transfer to other proteins, longer sequences or less skewed fitness distributions. Few draws. Five draws per cell give standard deviations that, at  $N \leq 96$ , are as large as most between-system differ-

ences; we report paired  $p$ -values descriptively and without multiple-comparison correction, and the pre-registered Wilcoxon test cannot reach the 0.05 level with five seeds. Tuning asymmetry. The CNN configuration was selected by hand on one separate draw at two sizes, evaluated on test variants that overlap with later evaluations' test pools; the linear models and the GP are tuned only on their training data. A fully nested protocol (inner validation for every model) was not run, and the CNN's raw-target advantage may partly reflect this. Ensembling was applied only in the CNN ablation. Reimplemented, simple baselines. The GP uses a plain Hamming kernel with a small hyperparameter grid, not structure- or language-model-informed kernels such as Kermut [12] or evolutionary density features [4]; the CNN is a tiny model on four residues, and no pretrained protein language model [2], [10] was used. Mechanisms are suggested, not isolated. The log-target ablation is consistent with the reading that the CNN's edge is in part an output nonlinearity (the gap narrowed but did not vanish at the largest size), but the log transform is a crude stand-in for a learned monotone function (a global-epistasis model [7] was not implemented), and the explanations offered for the pairwise-ridge collapse under dbl training and for the small- $N$  advantage of dbl training are hypotheses we did not test. Metrics. Top-100 recall is computed against the top 100 of the full test set (hundreds of thousands of variants) and is very sensitive to ties and to one or two variants at  $N \leq 384$ ; Spearman with massive ties at zero

TABLE IV  
Log-target ablation (rand and two dbl sizes): Spearman, mean  $\pm$  std over draws.

| Method                              | Task      | spearman $\uparrow$ | $n$ |
|-------------------------------------|-----------|---------------------|-----|
| One-hot ridge log target            | rand_48   | 0.26 $\pm$ 0.05     | 5   |
| GP log target                       | rand_48   | 0.27 $\pm$ 0.03     | 5   |
| One-hot + pairwise ridge log target | rand_48   | 0.27 $\pm$ 0.03     | 5   |
| One-hot ridge log target            | rand_96   | 0.37 $\pm$ 0.05     | 5   |
| GP log target                       | rand_96   | 0.37 $\pm$ 0.04     | 5   |
| One-hot + pairwise ridge log target | rand_96   | 0.38 $\pm$ 0.04     | 5   |
| One-hot ridge log target            | rand_384  | 0.52 $\pm$ 0.03     | 5   |
| GP log target                       | rand_384  | 0.53 $\pm$ 0.03     | 5   |
| One-hot + pairwise ridge log target | rand_384  | 0.52 $\pm$ 0.03     | 5   |
| One-hot ridge log target            | rand_2000 | 0.59 $\pm$ 0.01     | 5   |
| GP log target                       | rand_2000 | 0.59 $\pm$ 0.01     | 5   |
| One-hot + pairwise ridge log target | rand_2000 | 0.54 $\pm$ 0.01     | 5   |
| One-hot ridge log target            | dbl_384   | 0.46 $\pm$ 0.02     | 5   |
| GP log target                       | dbl_384   | 0.46 $\pm$ 0.03     | 5   |
| One-hot + pairwise ridge log target | dbl_384   | 0.44 $\pm$ 0.03     | 5   |
| One-hot ridge log target            | dbl_2000  | 0.50 $\pm$ 0.00     | 5   |
| GP log target                       | dbl_2000  | 0.48 $\pm$ 0.01     | 5   |
| One-hot + pairwise ridge log target | dbl_2000  | 0.45 $\pm$ 0.01     | 5   |

TABLE V  
Mean Spearman at rand<sub>384</sub> as a function of the fixed ridge penalty  $\alpha$ , and with the CV-selected penalty (main group).

| System         | $10^{-3}$ | $10^{-2}$ | $10^{-1}$ | 1     | 10    | $10^2$ | $10^3$ | CV    |
|----------------|-----------|-----------|-----------|-------|-------|--------|--------|-------|
| One-hot ridge  | 0.392     | 0.392     | 0.392     | 0.393 | 0.394 | 0.385  | 0.380  | 0.381 |
| Pairwise ridge | 0.348     | 0.348     | 0.351     | 0.371 | 0.399 | 0.389  | 0.382  | 0.396 |

fitness is dominated by the alive/dead split. What a full study would add. More landscapes (e.g. the benchmarks in ProteinGym [3]), many more draws, nested tuning for every model, a global-epistasis baseline, active learning rather than random draws, and language-model features.

## VIII. Conclusion

On the complete GB1 four-site landscape, with five draws per cell, we found no detectable benefit from pairwise epistasis features under random training (Spearman 0.478 versus 0.480 for the additive model at  $N=2000$ ) and a large loss when training on single and double mutants and testing on higher-order variants (0.196 versus 0.354). A Hamming-kernel GP matched additive ridge at small  $N$  and fell behind at  $N=2000$ . A small CNN had the best mean Spearman at  $N \geq 96$ , but a shared log-transformed target brought the linear models and the GP to within a narrow band of it at  $N=384$ , which is consistent with part of the CNN’s advantage on the raw target being a learned output nonlinearity (the gap narrowed but did not vanish at  $N=2000$ ). Of the five hypotheses, H5 is supported, H2 is supported in the mean but statistically inconclusive, and H1, H3 and H4 are refuted as stated (H4 only at  $N \leq 96$ ). These are small-scale results on one landscape with simple reimplemented baselines; they should be read as a calibrated reference point, not as a ranking of methods.

## References

[1] C. Dallago, J. Mou, K. E. Johnston, B. J. Wittmann, N. Bhattacharya, S. Goldman, A. Madani, and K. K. Yang, “FLIP: Benchmark tasks in fitness landscape inference for proteins,” 2021. [Online]. Available: <https://doi.org/10.1101/2021.11.09.467890>

[2] R. Rao, N. Bhattacharya, N. Thomas, Y. Duan, X. Chen, J. Canny, P. Abbeel, and Y. S. Song, “Evaluating Protein Transfer Learning with TAPE,” arXiv preprint arXiv:1906.08230, 2019. [Online]. Available: <https://arxiv.org/abs/1906.08230>

[3] P. Notin, A. W. Kollasch, D. Ritter, L. van Niekerk, S. Paul, H. Spinner, N. Rollins, A. Shaw, R. Weitzman, J. Frazer, M. Dias, D. Franceschi, R. Orenbuch, Y. Gal, and D. S. Marks, “ProteinGym: Large-Scale Benchmarks for Protein Design and Fitness Prediction,” 2023. [Online]. Available: <https://doi.org/10.1101/2023.12.07.570727>

[4] C. Hsu, H. Nisonoff, C. Fannjiang, and J. Listgarten, “Learning protein fitness models from evolutionary and assay-labeled data,” *Nature Biotechnology*, 2022. [Online]. Available: <https://doi.org/10.1038/s41587-021-01146-5>

[5] R. Michael, J. Kæstel-Hansen, P. M. Groth, S. Bartels, J. Salomon, P. Tian, N. S. Hatzakis, and W. Boomsma, “A systematic analysis of regression models for protein engineering,” *PLOS Computational Biology*, 2024. [Online]. Available: <https://doi.org/10.1371/journal.pcbi.1012061>

[6] N. C. Wu, L. Dai, C. A. Olson, J. O. Lloyd-Smith, and R. Sun, “Adaptation in protein fitness landscapes is facilitated by indirect paths,” *eLife*, 2016. [Online]. Available: <https://doi.org/10.7554/eLife.16965>

[7] J. Otwinowski, D. M. McCandlish, and J. B. Plotkin, “Inferring the shape of global epistasis,” *Proceedings of the National Academy of Sciences*, 2018. [Online]. Available: <https://doi.org/10.1073/pnas.1804015115>

[8] J. Zhou and D. M. McCandlish, “Minimum epistasis interpolation for sequence-function relationships,” *Nature Communications*, 2020. [Online]. Available: <https://doi.org/10.1038/s41467-020-15512-5>

[9] E. C. Alley, G. Khimulya, S. Biswas, M. AlQuraishi, and G. M. Church, “Unified rational protein engineering with sequence-based deep representation learning,” *Nature Methods*, 2019. [Online]. Available: <https://doi.org/10.1038/s41592-019-0598-1>

[10] J. Meier, R. Rao, R. Verkuil, J. Liu, T. Sercu, and A. Rives, “Language models enable zero-shot prediction of the effects of mutations on protein function,” 2021. [Online]. Available: <https://doi.org/10.1101/2021.07.09.450648>

[11] P. A. Romero, A. Krause, and F. H. Arnold, “Navigating the protein fitness landscape with Gaussian processes,” *Proceedings of the National Academy of Sciences*, 2013. [Online]. Available: <https://doi.org/10.1073/pnas.1215251110>

[12] P. M. Groth, M. H. Kern, L. Olsen, J. Salomon, and W. Boomsma, “Kermut: Composite kernel regression for protein variant effects,” 2024. [Online]. Available: <https://doi.org/10.1101/2024.05.28.596219>